Chapter 3

Endodontic problems in advanced operative dentistry

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CHAPTER CONTENTS

Endodontic health 33
General anatomical and physiological considerations 33
The dentine pulp complex 33
Nerves in the pulp 35
Pulp vasculature 35
Pulpal defence mechanisms 36
The effect of restorative procedures on the dental pulp 36
Health of the pulp 36
Preparation of the tooth 36
Frictional heat 36
The effects of local anaesthetic 37
The effects of dental materials on the dental pulp 37
Crown cementation 37
Marginal microleakage 37
Diagnosing the state of the dental pulp 38
Testing pulp sensitivity 38
Reliability of sensitivity testing 38
Tests using heat 39
Tests using cold 39
Electrical testing 39
The use of radiographs 40
Linking symptoms to pulpal status 40
Sharp pain 40
Lingering pain 40
Treatment for pulpal inflammation 41
Assessing teeth with pulp and periradicular disease 41
Radiographic assessment of the root-filled tooth 42
Survival of root-canal treated crowned teeth 43
Summary 44

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ENDODONTIC HEALTH

The health of the pulp and periradicular tissues is of paramount concern when undertaking operative and fixed prosthodontic procedures. Maintaining the health of the dental pulp is an ideal that should be realized if at all possible and consideration always given to the dental pulp when embarking upon operative care for a patient. Should the pulpal tissue become irreversibly damaged, then steps must be taken to prevent disease spreading to the periradicular tissues, or permit healing of those tissues if they are already diseased. Unfortunately, it is well known that operative procedures may cause damage to the pulp and this may be exacerbated by previous interventions. They may also cause disruption of the tooth surface, allowing subsequent entry of micro-organisms into the tooth. Pulpal disease is microbial in aetiology and it is important to prevent microbial elements reaching the dental pulp in such concentrations that irreversible damage occurs.

GENERAL ANATOMICAL AND PHYSIOLOGICAL CONSIDERATIONS

The dentine pulp complex

The dental pulp is encased within the hard tissue structure consisting of the enamel and dentine. As the dental pulp and the dentine are so inter-related it is better to consider

them together as the dentine–pulp complex and hence to consider dentine as a vital tissue.

The dentine is derived from complex cells, the odontoblasts, that form a layer at the periphery of the dental pulp (Figure 3.1). The dentine forms a tubular structure, each tubule filled with an odontoblastic process. The tubules are more highly concentrated at the pulp surface than at the enamel–dentine junction. There is a diffusion gradient across the dentine that is related to the number of tubules and the thickness of the dentine. The response of the pulp to progressive injury is by the deposition of dentine (reactionary or tertiary dentine) from the intact odontoblasts and also mineralization within the dentinal tubules (peritubular dentine), thereby reducing their diameter and their ability to allow diffusion through them (Figure 3.2). Thus, the tissue has modulating effects on restorative procedures and that, combined with inherent neutralizing capabilities of tissue fluids from the pulp for both acids and alkalis, means that the amount of residual dentine thickness, after tooth preparation, is inversely proportional to the damage that may occur to the pulp after chemical application to the dentine surface. This is important when considering the use of filling materials, including temporary dressings, and bonding techniques, where an acid is applied to the dentine surface.

Dentine is laid down throughout the life of the tooth: primary dentine as the tooth develops, secondary dentine after eruption of the tooth and tertiary dentine as a result of some insult to the tooth which may result in invasion of the dentine by bacteria or their toxic products. The aetiology of the last is usually dental caries or tooth wear, but may be induced by a restorative procedure (Figures 3.2 and 3.3). Odontoblasts are post-mitotic cells and may easily be damaged during restorative procedures. If this

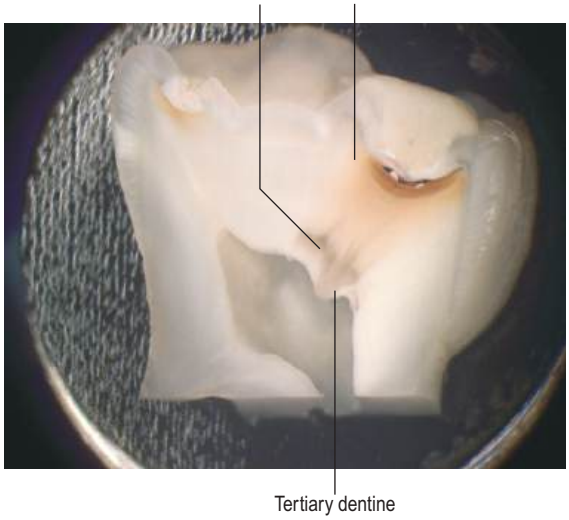


Figure 3.2 A hemisected tooth showing a carious lesion and dentine–pulp complex reactions.
(Courtesy of Dr Chris Longbottom)

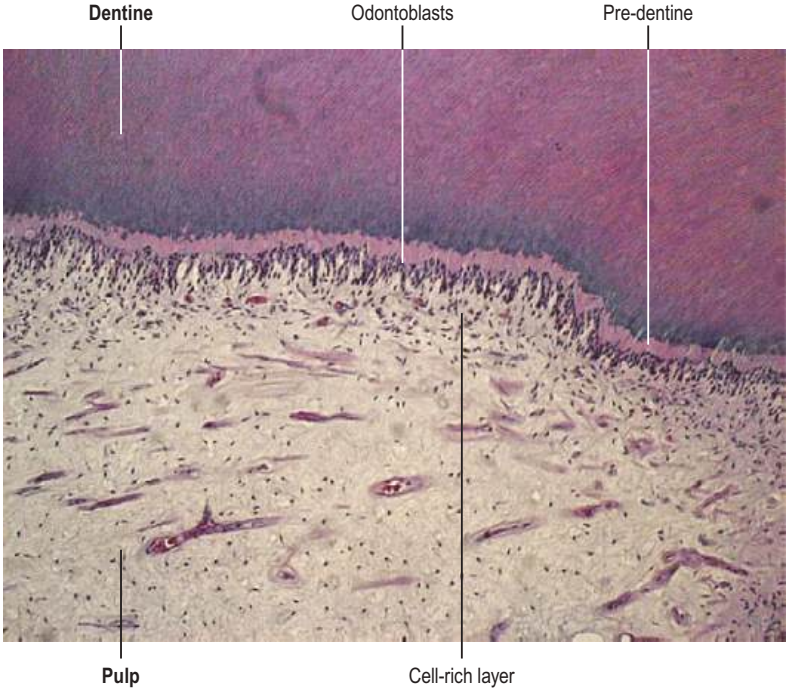


Figure 3.1 Decalcified section showing normal dentine pulp complex.

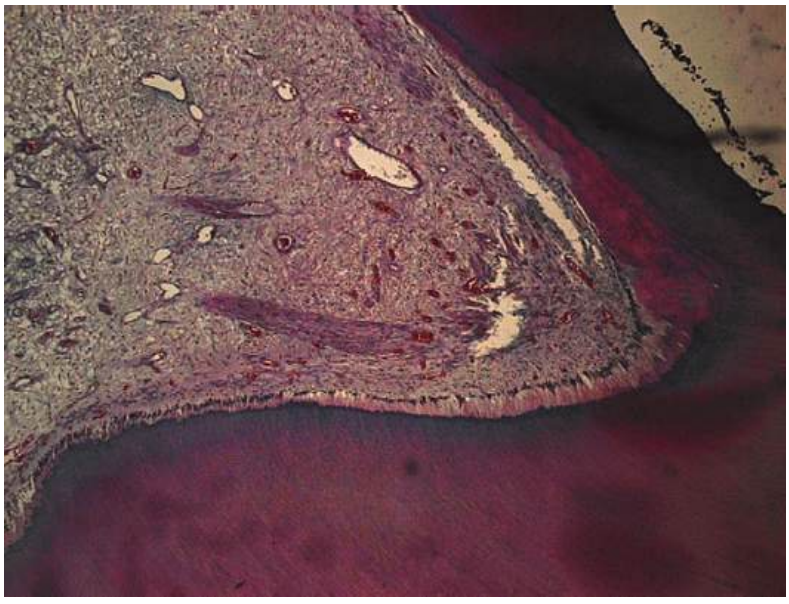


Figure 3.3 A decalcified section showing a carious lesion (top right corner) and tertiary dentine formation. The caries is likely to have been rapidly progressing as the tertiary dentine formation is less regular and has lost its tubular structure. Note the pulp is inflamed with an increased number of blood vessels and inflammatory cells.

occurs, they are replaced by stem cells from the pulp that differentiate into odontoblast-like cells and lay down mineralized tissue at the injury site. The quality of tertiary dentine is dependent upon the intensity of the insult to the pulp: in general terms, the greater the intensity of the insult (e.g. rapidly progressing caries), the more rapid the formation and the poorer the quality (Figure 3.3). Tertiary dentine is often porous, which makes it susceptible to invasion by micro-organisms and toxins. Therefore, once tertiary dentine is exposed by tooth wear or caries there is a greater risk of subsequent pulp pathology.

Nerves in the pulp

The principal elements of the dental pulp consist of stromal cells, fibroblasts, and an extensive network of nerves, blood vessels and lymphatics (see Figure 3.1). The nerves of the pulp conduct pain sensation. They consist of two main types: myelinated Aδ-fibres that conduct rapid, sharp pain sensation, and unmyelinated C-fibres that transmit the dull aching sensation that is typical of 'toothache'. In addition, nerves of the autonomic nervous system are also present. Various vasoactive neuropeptides are contained within these nerves. They are produced in the cell bodies of the trigeminal ganglion and are stored in the terminals of the nerves in the pulp. Release of these peptides causes a number of tissue reactions including changes in pulpal blood flow, blood vessel permeability and recruitment of immunocompetent cells as part of the inflammatory reaction. This is known as *neurogenic inflammation*, which may be damaging to the dental pulp. There are also effects on the growth of cells in the pulp, including fibroblasts and odontoblasts, where release of neuropeptides such as calcitonin gene-related peptide may cause the expression of bone morphogenetic protein-2 transcripts which results in more dentine formation. Recently it has been shown that fibroblasts in the pulp may also release these peptides, particularly substance P.

Pulp vasculature

The pulp shows a complex vascularity with a capillary bed adjacent to the odontoblast layer. The coronal part of the pulp has a more concentrated distribution of blood vessels than the radicular pulp. The pulp has a positive pressure compared with the outer surface of the tooth and this allows movement of fluid into the dentinal tubules. When the tubules are cut, and the tubule is patent, the outer cut surface of the dentine becomes wet as tissue fluid wells out. This provides a defensive mechanism, diluting toxic products that may enter the tubule and delivering immunocompetent cells and immunoglobulins to the surface. Unfortunately, this mechanism is usually insufficient to prevent penetration by micro-organisms, especially if the surface is heavily contaminated.

The dental pulp is often referred to as a low compliant tissue. This means that any increases in tissue pressure in the pulp – for example, by vasodilation of blood vessels during acute inflammation – may have devastating effects on the tissue because it is encased within a rigid hard tissue framework. It is essential that pathophysiological mechanisms are present to minimize such damaging effects on the pulp. Specific arteriovenous shunts may be opened to reduce damage to the capillary network and it is believed that lymphatic drainage has an important role in the reduction of tissue pressure.

Pulpal blood flow is higher than for other tissues in the oral region and is controlled by neuropeptide release and also the action of the autonomic nervous system. A transient increase in pulpal blood flow can be produced by various noxious stimuli when applied not just to the tooth in question but also adjacent teeth and adjacent tissues. This demonstrates the complexity of the nerve distribution in the oral cavity.

Pulpal defence mechanisms

The dental pulp is a complex structure that demonstrates various strategies to protect itself from injury. It has been shown to have good powers of healing through the deposition of tertiary dentine and mineralization of the peritubular dentine, even in the presence of bacterial contamination. The peritubular sclerosis gives rise to the histological appearance of translucent dentine (seen in ground sections; Figure 3.2) and the dentine in this region becomes more resistant to diffusion of bacteria and their noxious products. If, however, the damage is so severe that odontoblasts adjacent to the region of irritation die, then the tubules are left empty and are much less resistant to ingress of toxins and bacteria. These tubules are known as dead tracts.

Restorative intervention may cause damage to the pulp and it is important that the clinician is aware of this when undertaking operative procedures on the teeth and knows how best to minimize these effects.

THE EFFECT OF RESTORATIVE PROCEDURES ON THE DENTAL PULP

The effect that a restorative procedure has on the dental pulp is related to many factors. These are not only complex but also unpredictable, especially in terms of clinical outcome. They include the health of the pulp when the procedure is performed, the nature and amount of dentine remaining, the materials used for restoration, and the subsequent leakage of micro-organisms and toxins at the interface between the tooth and the restoration – a phenomenon termed microleakage.

Health of the pulp

Clinically it is difficult to tell the health of the dental pulp, especially if there are no outward clinical signs and symptoms. Previously restored teeth that require further clinical intervention may be at risk. Abou-Rass, in 1982, coined the term 'the stressed pulp'. This was a pulp that was neither healthy nor diseased but one that might be compromised to such an extent that further operative intervention would lead to irreversible damage. Taking a careful history from the patient regarding previous symptoms from, and treatment to, the tooth should be taken.

Nevertheless, the dental pulp may be injured during operative procedures on the tooth.

Preparation of the tooth

The most common method of tooth preparation is using rotating instruments in either an airotor, rotating at speeds up to 450 000 rpm, or in a slow speed handpiece that can be varied in speed from 1000 rpm to over 100 000 rpm. Cutting tooth tissue produces a smear layer, an amorphous mixture of inorganic and organic debris that blocks the cut dentinal tubules. Although the smear layer is inconsistent in thickness and distribution, it provides a barrier to bacteria, certainly in the short term. Many restorative procedures involve the removal of the smear layer, leaving open dentinal tubules which may make the pulp more vulnerable to damage by penetration of toxins and bacteria. Odontoblastic processes may also be cut during the preparation which may lead to subsequent irreversible damage to the pulp.

Frictional heat

The heat generated during tooth preparation for a restoration may cause irreversible damage to the pulp. The trauma damages the adjacent odontoblast layer and these cells can be sucked into their respective tubules. This is termed odontoblast aspiration and results in death of the cells and formation of a dead tract. These tracts are more susceptible to bacterial invasion as there is no protective outflow of tissue fluid or ability of the pulp tissue adjacent to the damaged area to form tertiary dentine quickly enough to prevent further damage to the pulp from the bacterial invasion. The microvasculature of the pulp is severely affected with plasma extravasation and a reduction in pulpal blood flow. In some cases the preparation may be so traumatic that haemorrhage occurs into the dental tubules and the pulp is irreversibly damaged immediately adjacent to the affected cut surface. If the tooth does not become non-vital then there may be an increased likelihood of internal root resorption taking place because of the local severe trauma.

Frictional heat occurs more readily with the following conditions:

- *Rotational speed on the bur*. The quicker the speed, the more damage is likely to take place.
- *Cooling the bur*. Cooling the bur, particularly with a jet of water, prevents damage to the pulp. The water must play on the whole cutting surface and for this reason a 'showerhead' water source from the handpiece is preferred. Most modern airotors have at least three water exits but it is essential that these are not blocked and sufficient water flow is present (Figure 3.4).
- *Efficiency of the bur*. A new bur should be used for the preparation of either intracoronal or extracoronal restorations. Diamond-coated burs become clogged with debris and it may be necessary to replace a bur



Figure 3.4 A high speed airotor with three jets of water directed to a tapered diamond bur. Good water coolant is necessary during tooth preparation.

during, for example, a full crown preparation. If the bur is blunt there is a tendency to use more force to achieve adequate cutting and the quality of modern handpieces is so good that stalling is unlikely to take place. This results in heat generation and injury.

- *Vibration*. It is believed that vibration of the tooth, especially at the frequencies generated by misaligned bearings in an airotor, may cause release of neuropeptides and induce neurogenic inflammation. It is important to check the handpiece to ensure the bur is running concentrically.

The effects of local anaesthetic

The use of laser Doppler flowmetry has shown that the delivery of local anaesthetic containing adrenaline adjacent to a tooth causes a reduction in or complete cessation of blood flow in the dental pulp. This may result in stagnation of the blood and a catastrophic fall in oxygen tension in the tissues. This does not result in irreversible damage to the pulp in most cases as the effect is present only for a short period during the duration of the local anaesthetic. However, the effects of overheating during tooth preparation are even more critical. In addition, multiple procedures on a tooth may place the pulp at risk. There is an argument to suggest that impressions using a rubber- or silicone-based material should not be undertaken at the same appointment that the preparation has been carried out on a tooth that has been anaesthetized in this way.

The effects of dental materials on the dental pulp

Following tooth preparation for indirect restorations, a temporary restoration is placed to protect the prepared tooth structure and prevent movement of the tooth. A number of temporary cements, especially those containing zinc oxide and eugenol, provide an obtundent effect on the pulp although eugenol has been shown to have toxic effects on the pulp, especially if placed in direct contact. The obtundent effect of eugenol on the dental pulp is attributed to its antinociceptive effects via the capsaicin receptor although there is some action on capsaicin-insensitive dental primary afferent neurones. In addition, eugenol has been shown to act like a local anaesthetic by inhibiting voltage-gated sodium channels and high-voltage activated calcium channel currents, and activating transient receptor potential vanilloid subtype 1. Eugenol also possesses antimicrobial properties; thus a zinc oxide–eugenol temporary dressing has sufficient free eugenol to prevent bacterial leakage. Some temporary filling materials consist of a paste-like composite-based material that hardens in the presence of fluid. This setting process may injure the underlying odontoblasts by dehydration and are thus not recommended in vital teeth.

Overall, the effects on the dental pulp of the most commonly used restorative materials is usually only mild and short-lived when there is no microbial contamination at the interface between the tooth and the material. There is evidence, however, that some inflammatory changes occur in the dental pulp in more than 50% of cases. Clinically, this reaction is probably not of consequence but it does add weight to the argument of the 'stressed pulp' condition.

Crown cementation

An informative review by Lam and Wilson (1999) concluded that sealing of the dentinal tubules after crown preparation and prior to cementation may be beneficial. Dentine bonding agents reduce dentinal fluid flow and therefore may prevent pulpal damage.

Marginal microleakage

Achieving a marginal seal at the interface between the restoration and the tooth has long been regarded as an important goal for the clinician. The presence of bacteria within this interface is associated with inflammation in the pulp. If this leakage is kept at a low level, the pulp responds defensively, with a reduction in dentine permeability and production of tertiary dentine. Nevertheless, much research has focussed on developing materials that bond to tooth structure in an attempt to reduce microleakage. Bonding to enamel is well known but there are concerns about long-term bonding to dentine. Nanoleakage arising from incomplete bonding in the hybrid layer allows penetration by fluids with subsequent degradation by hydrolysis. This will eventually allow bacterial penetration into the dentinal tubules and possible damage to the pulp.

It is important, therefore, to protect exposed dentine from bacterial penetration. The more dentine that becomes uncovered, the more dangerous it is, and the more likely

that damage to the pulp will occur. A preparation for a full crown is at much more risk than a shallow cavity in an occlusal pit or fissure.

It is clear that there are risks to the dental pulp during advanced operative procedures on teeth. This damage may manifest itself in the short term with the development of symptoms of acute pulpitis with sensitivity. This may be reversible or irreversible. The latter is particularly embarrassing for the practitioner as further, often complicated treatment is required, involving time and expense, on a tooth that was previously symptomless. Root canal treatment will be required, with access to the root canal system gained through the new restoration at best, but at worst following removal of the new restoration for improved access and vision.

There is evidence to show that advanced operative treatment may cause pulpal demise in the long term. A number of studies have shown the incidence of periradicular disease in crowned teeth not previously root-canal treated to be in the region of 10–18%. Most of these lesions are symptomless and are only discovered on incidental radiological examination.

Pulpal survival is linked to the type of restoration. The pulps in teeth with single ceramic fused to metal crowns perform better than those in teeth supporting a bridge retainer (Figure 3.5). Cheung et al. (2005) showed in their clinical study that the 10-year survival for the pulp was 84.4% for crowns and 70.8% for bridge retainers. This decreased to 81.2% for the crowns and 66.2% for the bridge retainers after 15 years. The reason for the difference between single units and bridgework may be the tendency to overcut teeth for bridge retainers to ensure parallelism. This causes more operative trauma to the tooth.

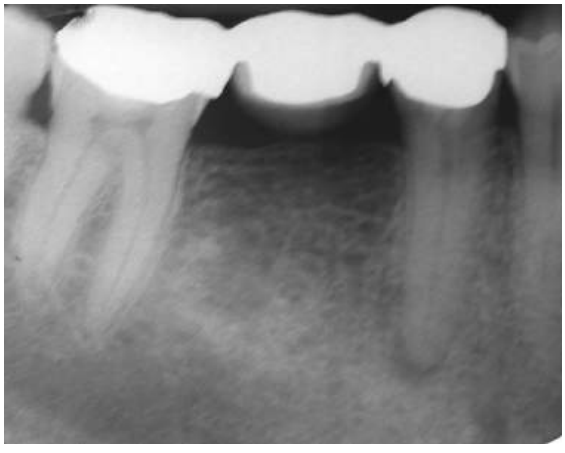


Figure 3.5 Periapical radiograph of lower right second premolar and molar teeth which have been prepared for a three unit fixed–fixed conventional bridge. The premolar tooth has a periradicular radiolucency and the distal root of the molar tooth is showing signs of root resorption.

It is important to note that the presence and quality of a coronal restoration is linked to the incidence of periradicular disease; teeth that are well restored are less likely to have periradicular disease because there is an improved coronal seal.

DIAGNOSING THE STATE OF THE DENTAL PULP

Testing pulp sensitivity

It is very difficult to diagnose the pathology of the dental pulp from diagnostic tests because direct inspection cannot be used. Indirect tests are usually employed and these generally test the sensitivity of the pulpal nerves using heat, cold (Figure 3.6) or electrical stimulation (Figure 3.7). This test is used to indicate pulpal necrosis. However, pulp vitality depends solely on blood supply and irreversible pulp damage may still be present despite a positive nerve response. More recently more sophisticated testing apparatus, using laser Doppler flowmetry or pulse oximetry, which determine blood flow and oxygen saturation levels, respectively, have also been used in research but no reliable product has been produced commercially for routine use in clinical practice.

Reliability of sensitivity testing

Although it is impossible to find a diagnostic test that provides 100% reliability all of the time, it is important to know how accurate a particular test might be. A useful study carried out by Petersson et al. and published in 1999, compared sensitivity testing with cold (ethyl chloride), heat (hot gutta-percha) and electricity (using a proprietary electric pulp tester). The results showed that the probability of obtaining a nil response in a case with a necrotic pulp was 89% with the cold test, 48% with the heat test and 88% with the electrical test. Conversely, the



Figure 3.6 Sensitivity testing with ethyl chloride.

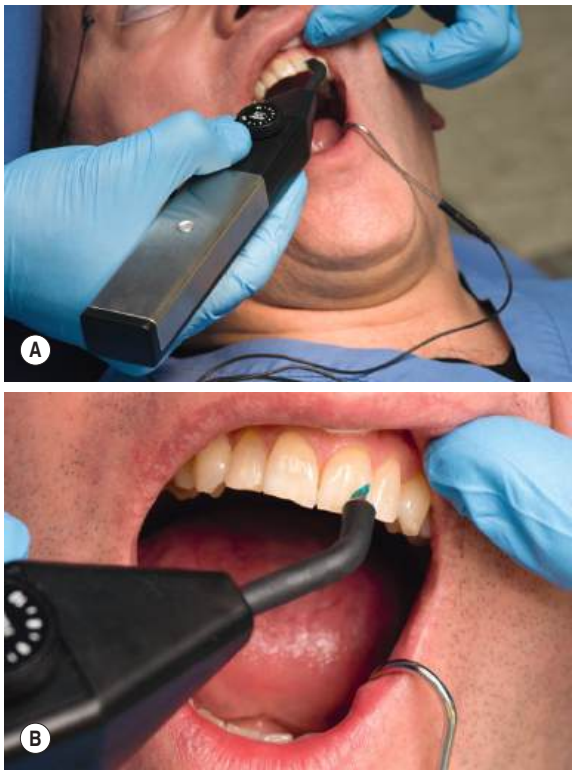


Figure 3.7 Sensitivity testing with an electrical pulp tester.

probability of a positive (sensitive) reaction representing a vital pulp was 90% with the cold test, 83% with the heat test and 84% with the electrical test. This study showed a relatively high probability that teeth not reacting to cold would have a necrotic pulp. In another study (Hyman & Cohen, 1984), however, this figure was less than a 50:50 probability. This may be because of the effect of having few teeth with extensive restorations and obliterated pulp chambers in the Petersson study. It is important to take note of how the presence of restorative materials may affect this test. A ceramic restoration provides excellent insulation but metal restorations are good conductors. Finally, the response from the patient is very subjective. For this reason it is best to test the contralateral tooth in addition to the tooth in question.

Tests using heat

A gutta-percha stick is heated and placed on the surface of the tooth. It is best to place a thin layer of petroleum jelly on the surface to prevent the gutta-percha sticking; it would be distressing for the patient if an extreme positive response was accompanied by the inability to remove the gutta-percha quickly from the tooth. In general, the heat test is not reliable and should be used only in combination with other tests.

Tests using cold

There are a number of ways cold can be applied to the tooth. Cold air, cold water or an ice stick affects fluid flow in the dentinal tubules and stimulates pulpal nerves. The most common method is the use of a cotton wool ball held in tweezers, sprayed with ethyl chloride (see Figure 3.6). As the chemical evaporates it cools rapidly to give a temperature of approximately –5°C. There is very little evidence to show that one material is better than another. When use of a dry ice stick has been compared with a refrigerant spray, 1,1,1,2-tetrafluoroethane, both have been found to be equivalent in eliciting a response, regardless of tooth type and presence of a restoration. However, the response with refrigerant spray is quicker. The use of refrigerant sprays is said to predispose to cracking of enamel because of the rapid fall in temperature, but there is no evidence for this. These sprays reduce the temperature to approximately –50°C.

Electrical testing

Proprietary electrical pulp testers are available. These send a weak electrical current through the tooth, stimulating the nerve and eliciting a pain response. There is a rheostat that allows an increasing current to be sent but there is no correlation between the reading and the state of the pulp (see Figure 3.7). Hyperaemic pulps may respond earlier than normal pulps. A bipolar device is presumed to be more accurate because the current is confined to the coronal pulp. Electrical pulp testing stimulates both Aδ and C fibres.

Clinical technique for electric pulp testing

The purpose of the test should be explained to the patient. The tooth should be dry and isolated with cotton wool rolls or a rubber dam to prevent salivary contamination. If an electrical device is being used, then a conducting medium should be placed on the probe tip (see Figure 3.7). This can be a fluoride gel or prophylactic paste. If possible, the application should be made at the incisal edge or as close to the pulp horn for maximum efficiency. Where the reaction is unclear, consider redoing the test with another device and testing teeth in a different order.

False-positive and false-negative responses to sensitivity testing can occur in a number of instances.

False-positive responses

- The use of sensitivity testing in multirooted teeth is unreliable, where a positive reaction may be elicited despite irreversible damage to the tooth. At least one root may contain vital tissue but the other roots may be necrotic.
- Improper technique (e.g. poor isolation). This could lead to contamination with superficial saliva and in the case of electric pulp testing, conduction of the electric current to an adjacent tooth or gingiva and stimulating these instead of the tooth being tested.

- An anxious patient may have a heightened perception or anticipation of a positive response.
- C-fibres may respond for a short period following loss of vitality.

False-negative responses

- Traumatized teeth. Following trauma a tooth may be concussed and not respond to sensitivity tests.
- Situations in which there is excess secondary and tertiary dentine formation (e.g. heavily restored teeth or teeth with extensive tooth wear).
- Patients with a high pain threshold.

The use of radiographs

The routine radiological image taken for examining the periradicular tissues of a tooth is the intraoral periapical radiograph. This should always be taken using a paralleling technique and with a beam-aiming device to minimize distortion of the resultant image. It is obvious that the health of the pulp cannot be seen on the radiograph; however, in a tooth with irreversible pulpitis, as the inflammation spreads through the root canal system, changes in the periradicular tissues lead to bone resorption. This is seen initially as loss of lamina dura. As the cortical plate is eroded, a radiolucency will be detected, but it may take up to 14 days for these changes to be seen radiographically. When the periradicular tissues are acutely inflamed, the tooth is usually tender to percussion and there may be tenderness on palpation in the buccal sulcus. Ultimately, if the tooth is untreated, the pulp may become necrotic and the periradicular lesion may become larger (Figure 3.8). At this stage there are usually no symptoms and a chronic periradicular periodontitis is diagnosed.



Figure 3.8 The crowned upper left central incisor has lost vitality and has a large periradicular radiolucency.

Linking symptoms to pulpal status

Symptoms of pain will also provide clues as to the state of the pulp but, unfortunately, these are not always reliable. Indeed, in the majority of cases, necrosis of the pulp occurs without symptoms. Conversely, a sharp pain that occurs spontaneously and perhaps disturbs sleep may not be easily attributed to a particular tooth, especially if a number of teeth on that side of the mouth are heavily restored or carious. There is some evidence to indicate that symptoms of pain increase as the histopathology of the dental pulp worsens. Patients presenting with intense or referred pain usually give a history of previous pain. The amount of pathology in the dental pulp can usually be determined by asking about a previous history of pain, and this may also help in determining the source of the pain in cases of referred pain.

Sharp pain

Sharp pain to cold stimuli that lasts only as long as the stimulus is present is linked to stimulation of Aδ-fibres. In these cases the pulp is usually reversibly inflamed. This pain is sometimes referred to as hypersensitivity and occurs when there are exposed dentinal tubules. Teeth affected by repeated bouts of these symptoms may develop an irreversibly damaged pulp. Hypersensitivity is related to teeth with exposed dentine and is typically seen in patients with the effects from erosion and abrasion and in teeth with gingival recession such as treated periodontal disease.

Lingering pain

Pain to hot stimuli, and occasionally cold, that lingers for minutes or hours after the stimulus is removed, is indicative of an irreversibly inflamed dental pulp, known as irreversible pulpitis (Figure 3.9). In these cases there is stimulation of the unmyelinated C-fibres, which generate neurogenic inflammation. Sometimes the pain is relieved by cold but again this is probably irreversible pulpitis.

Spontaneous pain which usually prevents the patient from sleeping is associated with severe inflammation of the pulp and irreversible pulpitis.

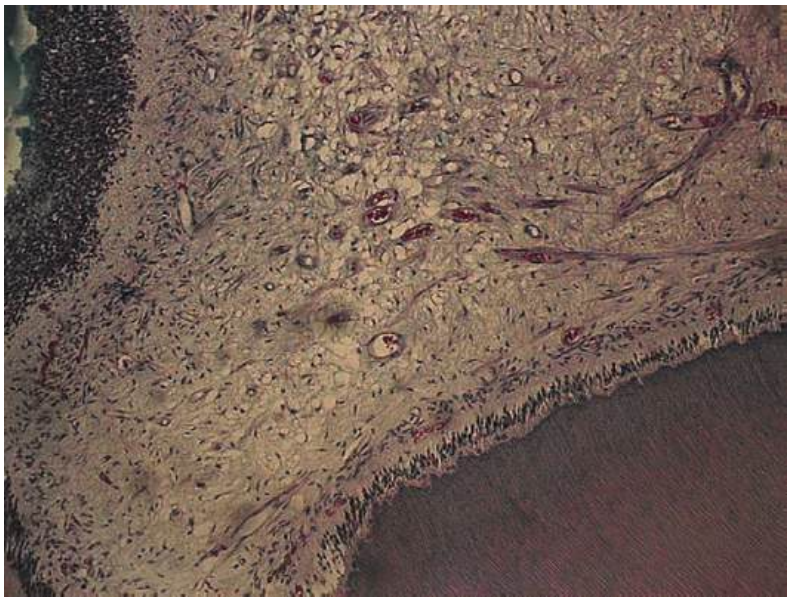


Figure 3.9 A decalcified section of a carious tooth which had symptoms of an irreversible pulpitis. Note the dense inflammatory infiltrate, top left.

Pain is a subjective emotion and pain thresholds vary from patient to patient. It is essential that a full history is taken, as well as appropriate special tests, prior to making a diagnosis of pulpitis. Acute symptomatic pulpitis progresses either to chronic pulpitis, which is usually painless, or to acute periradicular (periapical) periodontitis where the tooth becomes tender to tap, when an acute inflammatory reaction occurs in the periodontal ligament. Up until that time it may be difficult to diagnose which tooth is causing the problem.

TREATMENT FOR PULPAL INFLAMMATION

It is clear that the pulp has good reparative powers if the assault on the tissue is neither too severe nor prolonged. Teeth diagnosed with reversible pulpitis should have the pulp protected by blocking the tubule openings with a proprietary material. Those teeth with a clear irreversible pulpitis should be root-canal treated, or perhaps extracted if endodontics is not feasible, the tooth is unrestorable or is not strategic.

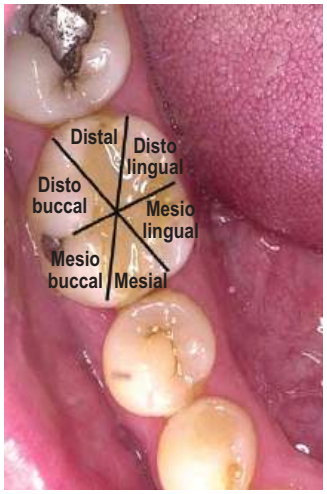
ASSESSING TEETH WITH PULP AND PERIRADICULAR DISEASE

Prior to treating irreversible pulpitis a clear indication of the quality of the remaining tooth structure must be made. As well as determining the presence of dental caries and the marginal integrity of the existing restoration, the clinician should carefully examine the tooth structure for cracks. This full examination cannot be done unless the restoration is removed (Figure 3.10). A study by Abbott (2004) showed that in 245 restored teeth from 220 patients referred for root canal treatment, 19.25% of the teeth had caries, 23.3% had cracks and 39.2% had marginal breakdown. However, when the restorations were removed, the incidence went up to 86.1%, 60% and 99.6%, respectively. Of these teeth, over 90% had more than one of these factors.

It is important to ascertain the extent of the remaining tooth tissue prior to embarking upon root canal treatment (Figure 3.10). It is embarrassing to undertake endodontic treatment and find that the tooth is unrestorable at the end of often complex, time-consuming and, inevitably, expensive treatment. Bandlish et al. (2006) have produced a useful system of measurement of remaining tooth structure (Figure 3.11). The crown of the tooth is divided into sextants and a score of 0–3 is given per sextant for the amount of tooth tissue remaining following preparation for the final restoration. Thus a maximum score of 18 is available for a wholly intact crown of the tooth. Consider the tooth in Figure 3.10; once the tooth has been root filled and the buccal cusp reduced for a full gold crown, for example, the restorability index will be low. It is not clear what minimum score is required before the tooth can be considered to be non-saveable and it also requires a vision of the tooth structure that might remain following final preparation. Further study of the longevity of restorations on root-filled teeth, applying these criteria, is required.



Figure 3.10 **A** The upper left first molar tooth is heavily restored and requires root canal treatment. It has an obvious crack palatally. The true extent of the crack can only be determined when the restoration has been removed. **B** Once the restoration was removed, the palatal cusp fractured. At this stage the restorability of the tooth can be assessed prior to carrying out root canal treatment.



0– **None**
Two thirds or more of sextant, no axial wall of dentin or no height of dentine to support restoration

1– **Inadequate**
Inadequate coronal dentin present but inadequate to contribute to retention (wall < 1.5 mm thick)

2– **Questionable**
More dentin than 1 but cannot be confident that it would contribute to retention/resistance

3– **Adequate**
There is sufficient coronal dentin in terms of height, thickness and distribution to contribute to retention/resistance

Maximum of 18 can be scored for tooth

Figure 3.11 Assessment of the amount of remaining coronal dentine in root-treated teeth.
(Reproduced with permission from Bandlish et al., 2006)

RADIOGRAPHIC ASSESSMENT OF THE ROOT-FILLED TOOTH

If a root-filled tooth is to be restored with an indirect restoration such as a crown, or is to be used as a bridge abutment tooth, a thorough evaluation of previous symptoms is required. A good quality periapical radiograph will provide the following important information:

- *Evidence of a periradicular radiolucency*. If possible, this radiograph should be compared with an image taken immediately after the root canal treatment was completed to compare changes.
- *Quality of the root canal treatment*. The radiograph will reveal the extent of the root filling, the density of the obturation and evidence of iatrogenic damage, including faults in the preparation, perforations and fractured instruments (Figure 3.12).
- *Presence of a post*. An indication of the type of post, its shape and extent can be ascertained (see Figure 3.8).

There is now some evidence to suggest that if a periradicular radiolucency in a root-filled tooth has not reduced in size within 1 year, then the treatment has failed and should be redone, especially if the tooth is to be restored with an extracoronal restoration. Scar tissue represents only about 2% of residual periradicular radiolucencies and can thus be discounted as a reason for the radiolucency. If the root filling is extruded through the apex of the tooth, or is more than 2 mm from the radiographic apex, then the success rate is reduced.

Periapical radiographs provide only a two-dimensional view of a three-dimensional structure. This limits diagnostic reliability, especially in multirooted teeth and in the presence of superimposed anatomical structures. In addition, periapical radiography has been shown to have low sensitivity in determining the presence of a periradicular radiolucency. The recent introduction of cone-beam computed tomography (cone-beam CT) has improved diagnostic capability in endodontic cases immeasurably. Cone-beam CT produces three-dimensional images of the site of interest and these images can then be manipulated using computer software. Periradicular lesions can be detected earlier and

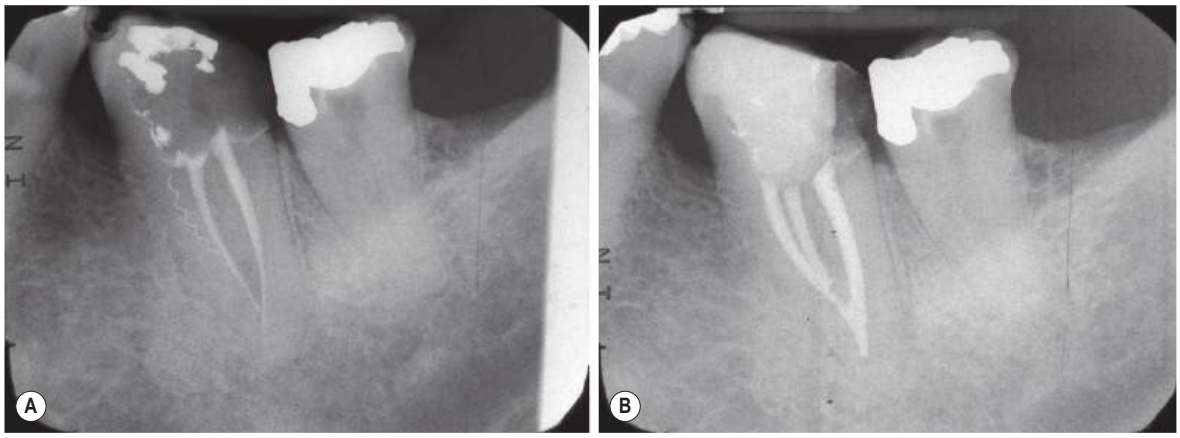


Figure 3.12 **A** Radiograph of lower second molar tooth showing fractured spiral filler in one of the two mesial canals. The obturation in the remaining canals appears to be of satisfactory length. **B** Radiograph of the tooth seen in **A**. The fractured instrument has been removed and root canal re-treatment carried out. The improved preparation and obturation clearly shows that the root canals fuse apically.

with greater consistency than with periapical radiography. Using the smallest field of view necessary it is possible to reduce radiation exposure to those of periapical radiography. Whilst cone-beam CT images are, at this stage, not recommended for all endodontic cases, they can be used in circumstances when diagnosis may otherwise be difficult or where non-surgical or surgical root canal treatment is being considered.

SURVIVAL OF ROOT-CANAL TREATED CROWNED TEETH

Patients may have concerns regarding the longevity of crowned teeth that have been root canal treated. For example, would it be more advantageous to have the tooth extracted and replaced by an implant-retained restoration? A systematic review of studies concerning the survival of root-filled teeth was undertaken by Stavropoulou and Koidis (2007). They were able to find 10 clinical studies that were suitable to be analysed. Root-canal treated teeth with a crown had a 10-year survival rate of 81%, compared with 63% for those root-canal treated teeth with a direct restoration. Other studies have also demonstrated that crowned teeth with good quality root fillings survive as well as crowned teeth that are vital.

So how might a root-canal treated tooth fail and why should carefully restoring such teeth be advantageous?

- *Structural integrity*. Root-canal treated teeth are, in general, compromised structurally. They often have lost a lot of tooth structure already and this, combined with cutting an access cavity and removing the roof of the pulp chamber, makes the tooth weaker. The preparation of the coronal part of the root canal system using rotary instruments, especially Gates Glidden and Peeso burs, may remove excessive amounts of tooth structure if not executed carefully.
- *Microcracks*. The dentine of root-canal treated teeth does not become harder, even 10 years after the treatment, but microcracks have been noted in the tooth structure of root-filled teeth. Occlusal forces may allow these cracks to propagate and lead to catastrophic failure of the tooth whilst in function (Figure 3.13).
- *Occlusal forces*. It has been shown that root-filled teeth are able to tolerate up to twice the occlusal load compared with vital teeth. This indicates that proprioceptive nerves fibres are almost certainly present in the pulp which sense occlusal forces that are applied and help prevent overload.
- *Chemical changes in the dentine*. It is known that some root canal sealers containing zinc oxide and eugenol have an effect on the physical properties of dentine. Zinc has been shown to migrate into dentine at the expense of calcium but the effect on the mechanical properties of the dentine is unknown. Eugenol also has an effect, increasing the hardness of the dentine but this reaches a maximum in 3 weeks.
- *Irrigants*. Irrigants used in root canal treatment also have been shown to have a negative effect on the mechanical properties of dentine. Sodium hypochlorite has been shown in some studies to have an effect and chelating agents such as citric acid and ethylene diamine tetra acetic acid (EDTA) remove the mineralized component of the dentine.



Figure 3.13 Vertical root fracture of a root filled molar tooth (arrowed). The loss of both mesial and distal marginal ridges, combined with removal of the roof of the pulp chamber, has weakened the tooth catastrophically.

- *Microbial action on dentine*. The interaction of microbes and food substrate, releasing acid, may also have an effect on the mechanical properties of the root dentine and, of course, dental caries at the root face will have a serious negative effect on tooth strength.
- *Microbial leakage*. Coronal leakage of micro-organisms, their by-products and nutrients has been shown to have a deleterious effect on the prognosis of root-filled teeth. All root canal filling materials leak and research has shown that gutta-percha, the most commonly used root canal filling material, does not resist leakage at its interface with the sealer. In addition, the dentinal tubules may be invaded by micro-organisms when the interface between the sealer and the root canal wall is disrupted. These bacteria then provide a reservoir for reinfection of the apical part of the root canal system which may cause the persistence or redevelopment of periradicular disease. Prevention or at least reduction of coronal leakage may be achieved by ensuring that a well-fitting, leak-proof restoration is placed after root canal treatment is completed. A base placed across the floor of the pulp chamber and the openings to the root canals has been shown to delay the initiation of periradicular disease and reduce the severity of inflammation in the periradicular tissues should periradicular periodontitis develop.

SUMMARY

It is clear that every effort should be made to prevent damage to the dental pulp during operative procedures. This relies, to a large extent, on the skill and knowledge of the practitioner. Care to protect the pulp during the preparation and subsequent restoration of the tooth is essential, bearing in mind the number of iatrogenic causes of pulp damage. This is particularly important in teeth that have been restored previously. Bacterial microleakage at the interface between the tooth and the restoration must be reduced and this is achieved by ensuring that the restoration adapts well to the tooth and, if possible, some micromechanical bonding of the restoration to the tooth structure is achieved, using specific techniques.

Should the pulp of the tooth become irreversibly damaged then the treatment options are either extraction or root canal treatment. The longevity of root-filled teeth depends on many factors but a proper restoration that protects both the remaining tooth structure and the root filling is essential.

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